

4) Navie Bayesian Classifier

```
import pandas as pd
import random as rd
from sklearn.metrics import confusion_matrix
import math

lut=dict()
label_attribute="PlayTennis"
labels=["No","Yes"]

df=pd.read_csv("/content/tennis.csv")
print(df)
total=len(df)

attribute_names=df.columns.tolist()
attribute_names.remove(label_attribute)
ldf=df.pivot_table(index=[label_attribute], columns=[label_attribute], aggfunc='size')
print(ldf)
lut[label_attribute]=ldf

for attribute in attribute_names:
    lut[attribute]=df.pivot_table(index=["PlayTennis"], columns=[attribute], aggfunc='size')
    lut[attribute].fillna(0,inplace=True)

target=list()
prediction=list()

for index, row in df.iterrows():
    if rd.random()>0.5:
        result=list()
        for label in labels:
            posteriorProb=math.log(lut[label_attribute][label][label]/total)
            for attribute in attribute_names:
                value=row[attribute]
                posteriorProb=posteriorProb+math.log(lut[attribute][value][label]/lut[label_attribute][label][label]+1)
            result.append(posteriorProb)

        target.append(row[label_attribute])
        print(result)
        maxindex=result.index(max(result))
        prediction.append(labels[maxindex])

print(confusion_matrix(target,prediction))
print(target)
print(prediction)
```

